

Sequences (S)

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June 20, 2026

Here are some powerful techniques in solving questions with sequences.

1 Linear recurrences

1.1 Linear Homogeneous Recurrences

Example 1: $a_{n+2} = 3a_{n+1} - 2a_n$, where $a_0 = 0, a_1 = 1$.

We start with the *ansatz* $a_n = x^n$. This gets us

$$x^{n+2} = 3x^{n+1} - 2x^n$$

$$x^2 - 3x + 2 = 0$$

$$(x - 1)(x - 2) = 0$$

Hence we conclude that $a_n = 2^n$ and $a_n = 1^n$ are solutions to the recurrence relation. Notice that any linear combination of these two solutions is also a solution to the recurrence. So the general solution to the recurrence is $a_n = A2^n + B1^n$ for some constants A and B . Since $a_0 = A + B = 0$ and $a_1 = 2A + B = 1$, we conclude that $A = 1, B = -1$. Hence, the solution to this recurrence is $a_n = 2^n - 1$.

Example 2: $a_{n+2} = 4a_{n+1} - 4a_n$, where $a_0 = 1, a_1 = 6$.

Using the same *ansatz*, we get the equation $(x - 2)^2 = 0$. Notice that we now have a root with multiplicity. So, alongside the usual solution of $a_n = 2^n$, we also have the solution $a_n = n2^n$ (If it was a triple root, then we have the solution $a_n = n^22^n$, etc (check it!)). As before, the general solution is $a_n = (A + Bn)2^n$ for some constants A, B , where we deduce $A = 1$ and $B = 2$ via the initial conditions. Thus, the solution is $a_n = (1 + 2n)2^n$.

Exercises: Find the general term.

1. $a_{n+3} = 6a_{n+2} - 11a_{n+1} + 6a_n$, where $(a_0, a_1, a_2) = (-1, 0, 4)$.
2. $a_{n+3} = a_{n+2} + a_{n+1} - a_n$, where $(a_0, a_1, a_2) = (2, 2, 6)$.
3. $F_{n+2} = F_{n+1} + F_n$, where $(F_0, F_1) = (0, 1)$.

Problems:

1. Find the general term of the fibonacci sequence $F_{n+2} = F_{n+1} + F_n, F_0 = 0, F_1 = 1$. Hence show that $F_{n+1}/F_n \rightarrow \phi$, where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.
2. Show that the general term of $a_{n+2} = a_{n+1} - a_n$, where $(a_0, a_1) = (0, 1)$ is $a_n = \frac{2}{\sqrt{3}} \sin\left(\frac{n\pi}{3}\right)$.

3. Let $\{a_n\}$ be a positive sequence such that $a_{n+2} = -a_{n+1} + 2a_n$. Show that a_n must be a constant sequence.
4. Let $\{a_n\}$ be an integer sequence such that $2a_{n+2} = 5a_{n+1} - 3a_n$. Show that a_n must be a constant sequence.
5. Let $\{a_n\}$ be a real valued sequence such that $a_{n+2} - ka_{n+1} + a_n = 0$, for some real number $|k| < 2$. Prove that a_n is bounded.
6. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $f(f(x)) = 4f(x) + 5x$.
7. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $f(x) > x$ and $f(f(x) - x) = 2x$.
8. (ISL 2010 A5) Find all functions $f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ such that $f(f(x)^2y) = x^3f(xy)$.
9. (China 2017) The sequences $\{u_n\}$ and $\{v_n\}$ are defined by $u_{n+2} = 2u_{n+1} - 3u_n$, where $(u_0, u_1) = (1, 1)$ and $v_{n+3} = v_{n+2} - 3v_{n+1} + 27v_n$, where $(v_0, v_1, v_2) = (a, b, c)$. There exists a positive integer N such that for any $n > N$, u_n divides v_n . Prove that $3a = 2b + c$.
10. (SMMC sample) Consider a circle with circumference $\frac{1+\sqrt{5}}{2}$ and mark a point P_0 on it. Going clockwise, you mark the points $P_1, P_2, \dots$ for every unit arc length. Suppose you terminate at step n , after you mark the point P_n . Show that if P_i, P_j are adjacent on the circle, then $|i - j|$ is a Fibonacci number.

1.2 Linear Inhomogeneous Recurrences

These are sequences with linear a_n terms, with along with some functions of n . The homogeneous version of the recurrence gives the general part of the solution, while you need a particular solution to take care of the inhomogeneous part (this is where you need to be smart).

Example 1: $a_{n+1} = 3a_n - 2$, where $a_0 = 0$.

The homogeneous version, $a_{n+1} = 3a_n$ has an easy general solution $a_n = A3^n$. We can guess a particular solution to the original equation, which is $a_n \equiv 1$. Thus, the general solution is $a_n = A3^n + 1$, where we tweak $A = -1$ to match the initial condition. Thus, the solution is $a_n = 1 - 3^n$.

The methodical way to do this is to rearrange the original equation to $a_{n+1} - 1 = 3(a_n - 1)$, and define $b_n = a_n - 1$. Then the b_n recurrence is $b_{n+1} = 3b_n$, etc. Sometimes guessing is not that easy, so these substitutions are useful.

Example 2: $a_{n+1} - 5a_n = 5^{n+1}$, where $a_0 = 0$

Divide both sides with 5^{n+1} to get $\frac{a_{n+1}}{5^{n+1}} - \frac{a_n}{5^n} = 1$. Define $b_n = \frac{a_n}{5^n}$, then we get $b_{n+1} - b_n = 1$, with $b_0 = \frac{a_0}{5^0} = 0$. This yields $b_n = n$, so we conclude $a_n = n5^n$.

Problems:

1. $a_{n+1} - 5a_n = (2n + 1)5^{n+1}$, where $a_0 = 0$.
2. (AMO 2015) Define the sequence $\{a_n\}$ such that $a_{n+2} = 2a_{n+1} - a_n + 2$ with $(a_0, a_1) = (4, 7)$. Show that $a_n a_{n+1}$ is always a term of the sequence.
3. (102 PiC) How many subsets of $\{1, 2, 3, \dots, 2020\}$ has the property that the sum of its elements is a multiple of 5?

1.3 First Order Multi-variable Homogeneous Recurrences

Sometimes you have more than one sequence working in tandem to define each other.

Example: Find the number of binary sequences of length n such that there does not exist consecutive zeros.

Let a_n, b_n be the number of binary sequences of length n ending with 0 and 1 respectively. As there cannot be consecutive zeros, we have $a_{n+1} = b_n$ (every sequence counted in a_{n+1} is a 0 subtended to a b_n sequence) and $b_{n+1} = a_n + b_n$ (every sequence counted in b_{n+1} is 1 subtended to a a_n or b_n sequence).

Since we're only interested in the sum $x_n := a_n + b_n$, let us inspect this instead.

$$x_n = a_n + b_n = b_{n-1} + b_n = (a_{n-2} + b_{n-2}) + (a_{n-1} + b_{n-1}) = x_{n-2} + x_{n-1}$$

Hence the recurrence is the Fibonacci recurrence.

What if the double-recurrence isn't nice enough so that such manipulations are not immediately obvious?

Our double recurrence can be expressed as $\begin{pmatrix} a_{n+1} \\ b_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_n \\ b_n \end{pmatrix}$. Let $\mathbf{x}_n := \begin{pmatrix} a_n \\ b_n \end{pmatrix}$ and $A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. Then $\mathbf{x}_{n+1} = A\mathbf{x}_n$. Thus, our general form becomes $\mathbf{x}_n = A^n \mathbf{x}_0$. The real task is figuring out how to find A^n in a reasonable amount of time. You can *diagonalise* the matrix such that $A = PDP^{-1}$ for some diagonal matrix D (which means we can find D^n easily), so we get $A^n = PD^nP^{-1}$, which is actually solvable (details not included for sanity; look up how to diagonalise a matrix).

1.4 Pell equations

In number theory, this is a Pell equation: $x^2 - dy^2 = 1$ for a non-square number d .

Example: $x^2 - 2y^2 = 1$.

We need an initial solution. There is no method to find this, so good luck with your guesses. In this case, $(x, y) = (3, 2)$ is a solution. Let (x_n, y_n) be a solution to the equation. Then we do a weird factorisation.

$$\begin{aligned} (x_n - y_n\sqrt{2})(x_n + y_n\sqrt{2}) &= 1 \\ (3 - 2\sqrt{2})(3 + 2\sqrt{2}) &= 1 \end{aligned}$$

Multiplying the terms at their respective locations,

$$((3x_n + 4y_n) - (2x_n + 3y_n)\sqrt{2})((3x_n + 4y_n) + (2x_n + 3y_n)\sqrt{2}) = 1$$

So we see that $(x_{n+1}, y_{n+1}) = (3x_n + 4y_n, 2x_n + 3y_n)$ is a new solution to this Pell equation.

Before you start to write matrices, look a bit closer on how we generated the solutions. $x_n \pm y_n\sqrt{2} = (3 \pm 2\sqrt{2})^n$.

$$x_n = \frac{(3 + 2\sqrt{2})^n + (3 - 2\sqrt{2})^n}{2} \quad y_n = \frac{(3 + 2\sqrt{2})^n - (3 - 2\sqrt{2})^n}{2\sqrt{2}}$$

Problems:

- (AMO 2020) Sequences $\{a_n\}$ and $\{b_n\}$ satisfy $a_1 = b_1 = 1$ and $a_{n+1} = \frac{a_n+2}{a_n+1}$ and $b_{n+1} = \frac{b_n}{2} + \frac{1}{b_n}$. Show that $b_{n+1} = a_{2^n}$.
- (Crux) $x_{n+2} = x_{n+1}\sqrt{x_n^2 + 1} + x_n\sqrt{x_{n+1}^2 + 1}$ satisfies $x_0 = x_1 = 1$. Find its general form.
- (Jongmin) Positive integers x, y satisfy $x^2 = (3y + 4)(y + 1)$. Find all solutions.

2 Telescoping

We want to turn the summation or product into the following forms to simplify nicely.

$$\sum_{k=1}^n a_k - a_{k-1} = a_n - a_0$$

$$\prod_{k=1}^n \frac{a_k}{a_{k-1}} = \frac{a_n}{a_0}$$

Exercises:

1. Simplify $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \cdots + \frac{1}{9999 \times 10000}$
2. Simplify $\sum_{k=0}^{2017} \frac{k}{(k+1)!}$
3. Simplify $\sum_{k=0}^{2017} \binom{k+8}{k}$
4. Simplify $\prod_{k=1}^{2017} \frac{k^2 + 5k + 4}{k^2 + 5k + 6}$
5. Prove that $1.5 < \sum_{n=1}^{\infty} \frac{1}{n^2} < 2$

Problems:

1. (Balkan 2016) Find all injective functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that for $x \in \mathbb{R}$ and $n \in \mathbb{Z}^+$,

$$\left| \sum_{i=1}^n i (f(x+i+1) - f(f(x+i))) \right| < 2016$$

2. (ISL 2015) Suppose that a sequence $a_1, a_2, \dots$ of positive real numbers satisfies

$$a_{k+1} \geq \frac{ka_k}{a_k^2 + (k-1)}$$

for every positive integer k . Prove that $a_1 + a_2 + \dots + a_n \geq n$ for every $n \geq 2$.

3. (USAMTS) Determine the value of

$$S = \sum_{k=1}^{2016} \sqrt{1 + \frac{1}{k^2} + \frac{1}{(k+1)^2}}$$

3 Convergence

Completeness axiom of real numbers: Every subset $S \subseteq \mathbb{R}$ with an upper bound has a *least upper bound* in $\mathbb{R}$, also known as the supremum s . Note that this axiom does not necessitate that $s \in S$.

Exercises:

1. Let $A = \{n \in \mathbb{Z}^+ \mid 1 - 2^{-n}\}$. What is the supremum s of A ? Is s an element of A ?
2. Show that every subset $S \subseteq \mathbb{R}$ with a lower bound has a *greatest lower bound* in $\mathbb{R}$, also known as the infimum.
3. Show that this property does not hold for $\mathbb{Q}$. Can you think of an example of a bounded subset of $\mathbb{Q}$ that does not have a supremum in $\mathbb{Q}$?
4. Show that every monotonically increasing sequence with an upper bound converges to a finite limit.
5. Show that every monotonically decreasing sequence with a lower bound converges to a finite limit.
6. Let $a_0 = 2$ and $a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}$. Show that a_n converges. What is the limit?

Problems:

1. (Jongmin) Consider a sequence given by $a_{n+1} = \sqrt{2^{a_n}}$, $a_0 = 1$.
 - (a) Show that $\{a_n\}$ converges. What is the limit? Why is the limit not 4?
 - (b) For which values of a_0 does $\{a_n\}$ converge? What number does it converge to for each value of a_0 ?
2. (AMO 1989) u_n satisfies $u_1 = 1$, and $u_1 + u_2 + \dots + u_n = \frac{1}{u_{n+1}}$. Determine whether there exist m such that

$$\sum_{i=1}^m u_i \geq 1989$$

3. (SMMC 2017) Let $a_1, a_2, a_3 \dots$ be the sequence of real numbers defined by $a_1 = 1$ and for $m \geq 2$

$$a_m = \frac{1}{a_1^2 + a_2^2 + \dots + a_{m-1}^2}$$

Show whether there exist N such that

$$a_1 + a_2 + \dots + a_N > 2017^{2017}$$

4. (CMC 2020) Let $a_1, a_2, a_3 \dots$ be the sequence of positive real numbers such that for each $n \in \mathbb{Z}^+$,

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_{n+1}^2}{n+1}}$$

Show that $\{a_n\}$ is a constant sequence.

4 Generating functions

Given a sequence $\{a_n\}$, we consider a formal power series $f(x) = a_0 + a_1x + a_2x^2 + \dots$. We call this the generating function corresponding to the sequence $\{a_n\}$. This is useful as we will see very soon.

4.1 How to get generating functions from sequences

Example 1: What is the generating function of $a_n = 2^n$?

Indeed the generating function is $f(x) = 1 + 2x + 4x^2 + \dots$. For generating function purposes, we don't really care about the technicalities of convergence. We just use the geometric series formula to simplify $f(x) = \frac{1}{1-2x}$.

Example 2: What is the generating function of $a_n = n$?

Start with the generating function $f(x) = \frac{1}{1-x} = 1 + x + x^2 + \dots$. We can differentiate both sides to obtain

$$f'(x) = \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots = \sum_{n=0}^{\infty} (n+1)x^n$$

However notice that our terms are off by one (we want $\sum_{n=0}^{\infty} nx^n$). We can fix this easily by multiplying x to both sides. Thus the generating function is $xf'(x) = \frac{x}{(1-x)^2}$.

Example 3: What is the generating function of $a_{n+2} = 3a_{n+1} - 2a_n$ with $a_0 = 2$ and $a_1 = 3$?

Let $f(x) = a_0 + a_1x + a_2x^2 + \dots$ be the generating function. Then notice

$$\begin{aligned} f(x) &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots + a_{n+2}x^{n+2} + \dots \\ xf(x) &= a_0x + a_1x^2 + a_2x^3 + a_3x^4 + \dots + a_{n+1}x^{n+2} + \dots \\ x^2f(x) &= a_0x^2 + a_1x^3 + a_2x^4 + \dots + a_nx^{n+2} + \dots \end{aligned}$$

Thus according to the formula, most of the terms will cancel out in $f(x) - 3xf(x) + 2x^2f(x)$. Let's see what terms are left.

$$f(x) - 3xf(x) + 2x^2f(x) = a_0 + a_1x - 3a_0x = 2 - 3x$$

Thus the generating function is $f(x) = \frac{2-3x}{1-3x+2x^2}$. However, notice the following

$$\begin{aligned} f(x) &= \frac{2-3x}{1-3x+2x^2} \\ &= \frac{2-3x}{(1-x)(1-2x)} \\ &= \frac{1}{1-x} + \frac{1}{1-2x} \\ &= (1 + x + x^2 + \dots) + (1 + 2x + 4x^2 + \dots) \\ &= \sum_{n=0}^{\infty} (1 + 2^n)x^n \end{aligned}$$

Which is consistent with our understanding of this sequence, because the general term is indeed $a_n = 1 + 2^n$.

Exercises:

1. What is the generating function for $a_n = n^2$?
2. What is the generating function for $a_n = n2^n$?
3. What is the generating function for the Fibonacci sequence given by $F_{n+2} = F_{n+1} + F_n$ and $F_0 = F_1 = 1$?

4.2 How to get sequences back from generating functions

Example 1: What is the sequence defined by the generating function $f(x) = e^x$?

We use the Taylor series. The $n!a_n$ would be the constant term after differentiating n times.

$$a_n = \frac{1}{n!} \left[\frac{d^n}{dx^n} f(x) \right]_{x=0} = \frac{1}{n!} [e^x]_{x=0} = \frac{1}{n!}$$

Thus $f(x)$ defines the sequences $a_n = \frac{1}{n!}$.

Exercises:

1. What is the sequence given by $f(x) = \sqrt{1+x}$?
2. What is the sequence given by $f(x) = \ln(x+1)$?

4.3 How to use generating functions to count things

Consider a_n as the number of ways to choose n things satisfying some condition, with power series $f(x) = a_0 + a_1x + a_2x^2 + \dots$. Similarly assume we have another sequence $\{b_n\}$ and the power series $g(x) = b_0 + b_1x + b_2x^2 + \dots$, and say b_n is the number of ways to choose n things satisfying another condition. One might ask, how many ways can we choose n things that satisfies both conditions? We can choose k things satisfying the first condition, and $n-k$ things satisfying the second condition, and add it all up for $k = 0$ to $k = n$. If we call this c_n , then we have

$$c_n = a_0b_n + a_1b_{n-1} + \dots + a_{n-1}b_1 + a_nb_0$$

Coincidentally, this is exactly what happens to polynomial coefficients when you multiply them (wow!). Thus our new generating function is $h(x) = c_0 + c_1x + c_2x^2 + \dots = f(x)g(x)$.

Example 1: You have 1 apples, 2 bananas, 3 carrots. How many ways can you choose 4 things?

The gf for apples is $a(x) = 1+x$, the gf for bananas is $b(x) = 1+x+x^2$ and the gf for carrots is $c(x) = 1+x+x^2+x^3$. The gf for the whole thing is $a(x)b(x)c(x) = (1+x)(1+x+x^2)(1+x+x^2+x^3) = x^6 + 3x^5 + 5x^4 + 6x^3 + 5x^2 + 3x + 1$. Since the coefficient for x^4 is 5, the answer is 5.

Example 2: How many ways can you choose k people out of a class of n people?

The gf for a single person is $1+x$, since there's one way to choose zero people out of this one person, and one way to choose one person out of this person (weird phrasing I know). Now, the gf for n people is $(1+x)(1+x)\dots(1+x) = (1+x)^n$. We all know that the coefficient of the n -th term is $\binom{n}{k}$ which is indeed the answer.

Example 3: At a supermarket, there are n different items. How many ways can you choose k things allowing repetition?

The gf for a single item is $\frac{1}{1-x} = 1 + x + x^2 + \dots$, since there's one way to choose zero things, one way to choose one thing, one way to choose two things, etc. Now, the gf for n items is $\frac{1}{(1-x)^n}$. We can obtain the sequence back

from the generating function by using the following observation

$$\begin{aligned}
 (n-1)! \frac{1}{(1-x)^n} &= \frac{d^{n-1}}{dx^{n-1}} \frac{1}{1-x} \\
 (n-1)! \frac{1}{(1-x)^n} &= \frac{d^{n-1}}{dx^{n-1}} \sum_{k=0}^{\infty} x^k \\
 (n-1)! \frac{1}{(1-x)^n} &= \sum_{k=0}^{\infty} (n+k-1)(n+k-2) \cdots (k+1) x^k \\
 \frac{1}{(1-x)^n} &= \sum_{k=0}^{\infty} \frac{(n+k-1)(n+k-2) \cdots (k+1)}{(n-1)!} x^k
 \end{aligned}$$

Thus we retrieve the formula for combination with repetition $\binom{n+k-1}{k}$.

Example 4: Let the Fibonacci sequence be given by $F_{n+2} = F_{n+1} + F_n$ and $F_0 = F_1 = 1$. What is $\sum_{n=0}^{\infty} F_n 2^{-n}$? Sometimes when we know the power series converges, we can calculate interesting sums. We can find that the generating function is

$$F(x) = F_0 + F_1 x + F_2 x^2 + \cdots = \frac{1}{1-x-x^2}$$

Subbing in $x = \frac{1}{2}$, we see that $\sum_{n=0}^{\infty} F_n 2^{-n} = F(\frac{1}{2}) = 4$.

Exercises:

1. You have apples, bananas, carrots, and durians. You have to choose a multiple of 5 apples, even number of bananas, at most 1 carrot and at most 4 durians. How many ways can you choose 2020 things?

Problems:

1. Consider an $n \times n$ grid. Let c_n be the number of shortest paths along the gridlines from the bottom left corner to the top right corner. Find the general form for c_n . What is $\sum_{n=0}^{\infty} \frac{c_n}{4^n}$?
2. An alternating permutation is a permutation of $\{1, 2, \dots, n\}$ such that $a_1 < a_2 > a_3 < a_4 > a_5 \cdots$. How many alternating permutations are there?
3. (SMMC 2018) For each positive integer n , consider a cinema with n seats in a row, numbered left to right from 1 up to n . There is a cup holder between any two adjacent seats and there is a cup holder on the right of seat n . So seat 1 is next to one cup holder, while every other seat is next to two cup holders. There are n people, each holding a drink, waiting in line to sit down. In turn, each person chooses an available seat uniformly at random and does the following:
 - (a) If they sit next to two empty cup holders, then they choose a cup holder at random and put their drink in it
 - (b) If they sit next to one empty cup holder, then they place their drink in that cup holder.
 - (c) Otherwise, they get angry.

Let p_n be the probability that no one gets angry. What is $p_1 + p_2 + p_3 + \cdots$?